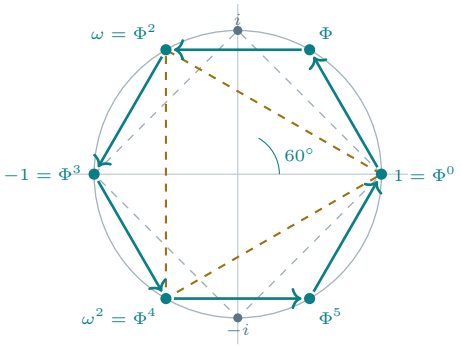
ALGEBRA ON ONE PAGE

Euler's formula: rotating round the polygons

Multiplying by $e^{i\theta}$ rotates by θ, so powers of i, ω, Φ step round polygons, and i^i turns a rotation into a shrink.

1 / THE RULE AND THE PICTURE

Euler's formula turns multiplication into rotation



one step $\times\Phi$; dashed triangle: $\times\omega = \Phi^2$; grey square: $\times i$

For every angle θ, the point of the unit circle at angle θ is

$$\boxed{e^{i\theta} = \cos\theta + i\sin\theta}$$

and since $e^{i\alpha}e^{i\beta} = e^{i(\alpha+\beta)}$, multiplying by $e^{i\theta}$ rotates by θ. The nth roots of unity are the corners $\zeta_n^k = e^{2\pi ik/n}$ of a regular n-gon, and one step round the polygon is multiplication by ζ_n. So $\Phi = e^{i\pi/3}$ steps round the hexagon, $\omega = \Phi^2$ takes two steps at a time and visits the triangle, and $i = e^{i\pi/2}$ steps round the square. Taking $\theta = \pi$ gives $e^{i\pi} = -1$.

2 / ONE STEP ROUND EACH POLYGON

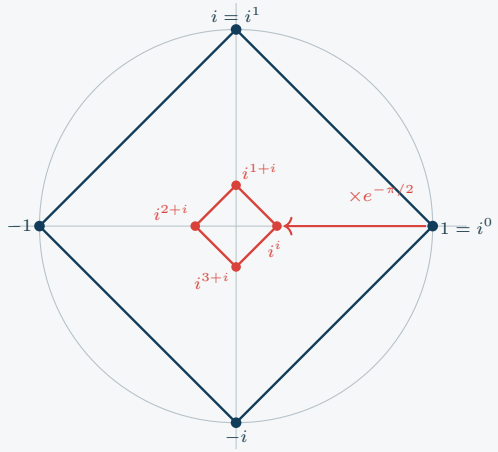
The symbols of the previous sheet are all of the form $e^{i\theta}$

symbol	$e^{i\theta}$	angle θ	polygon	steps to return
i	$e^{i\pi/2}$	$90°$	square	4
ω	$e^{2\pi i/3}$	$120°$	triangle	3
Φ	$e^{i\pi/3}$	$60°$	hexagon	6
ζ_{12}	$e^{i\pi/6}$	$30°$	12-gon	12
-1	$e^{i\pi}$	$180°$	two corners	2

Returning to 1 after the stated number of steps is the statement $e^{2\pi i} = 1$, and the powers agree with the corner labels: $i^2 = -1$, $\Phi^2 = \omega$, $\Phi^3 = -1$, $\omega^3 = 1$.

3 / A COMPLEX EXPONENT TURNS AND SHRINKS

The example i^i: the square of fourth roots, scaled by $e^{-\pi/2}$



A power with a complex exponent is defined through e, and $i = e^{i\pi/2}$ gives $i^z = e^{z\,i\pi/2}$. For $z = a + bi$ the exponent is $-\frac{\pi b}{2} + \frac{i\pi a}{2}$, so

$$\boxed{i^{\,a+bi} = e^{-\pi b/2}\cdot e^{i\pi a/2}}$$

The real part a counts quarter turns round the square and the imaginary part b scales by $e^{-\pi b/2}$. With $a = 0$ and $b = 1$ there is no turn and

$$i^i = e^{i\cdot i\pi/2} = e^{-\pi/2} \approx 0.20788,$$

a real number.
The four values $i^{k+i} = i^k\, i^i$ are the corners of the same square shrunk by 0.208, as drawn. Since $i = e^{i(\pi/2+2\pi k)}$ for every whole k, a full extra lap round the square gives further values $i^i = e^{-\pi/2-2\pi k}$, namely $\ldots, 111.3,\ 0.2079,\ 0.000388, \ldots$, each a factor $e^{2\pi} \approx 535.49$ from the last.

4 / THE SAME IDEA FOR EVERY POLYGON

Putting i into the exponent trades the rotation angle for a shrink factor

polygon n	$\zeta_n^{\,i} = e^{-2\pi/n}$	value
2	$(-1)^i = e^{-\pi}$	0.0432
3	$\omega^i = e^{-2\pi/3}$	0.1231
4	$i^i = e^{-\pi/2}$	0.2079
6	$\Phi^i = e^{-\pi/3}$	0.3509
12	$\zeta_{12}^{\,i} = e^{-\pi/6}$	0.5924

In general $\zeta_n^{\,i} = e^{i\cdot 2\pi i/n} = e^{-2\pi/n}$, so the angle $\frac{2\pi}{n}$ of one step becomes the factor $e^{-2\pi/n}$. More corners mean smaller steps and less shrinking, and the factor tends to 1 as n grows. This is the flip of the golden-metallic sheet, $t \leftrightarrow i\theta$: circular rotation on one side, real growth or decay on the other.

ASIDE / TRY THESE

Show $\Phi^6 = e^{2\pi i} = 1$. Find ω^i: $e^{-2\pi/3} \approx 0.1231$. What is i^{-i}? It is $e^{\pi/2} \approx 4.81$.

LINKS TO OTHER SHEETS

The roots of unity of order $3, 4, 6, 12$ sheet; the golden and metallic sides sheet, where $t \leftrightarrow i\theta$.

Convention. $e^{i\theta} = \cos\theta + i\sin\theta$; $\zeta_n = e^{2\pi i/n}$; principal values use $i = e^{i\pi/2}$, further values $i = e^{i(\pi/2+2\pi k)}$; numerical values were checked.

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